

## CASE STUDY

# SpecCopilot — an AI tool that turns a one-line idea into a full PRD

Vimal Ajit · built solo, end-to-end · live demo: [vimalajit11.github.io/spec-copilot](https://vimalajit11.github.io/spec-copilot)

### The problem

Drafting a PRD from a blank page is one of the most repetitive, time-consuming parts of the PM job — and the part most amenable to AI. The bottleneck isn't the thinking; it's the cold-start tax of turning a one-line idea into structured problem framing, user stories, acceptance criteria, metrics, and a prioritization call. I built SpecCopilot to collapse that phase, as a working example of the kind of internal productivity tool that multiplies a whole product team.

### What I built

A single-file web app that takes a rough feature idea and product context, calls the Claude API, and returns a structured PRD live: problem, objective, target users, user stories, testable acceptance criteria, quantified success metrics, risks, and a RICE score with a visual breakdown. The output is one click from export as clean Markdown, so it drops straight into Notion, Confluence, or Jira where PRDs actually live.

### How I leveraged AI — the decisions, not just the prompt

- **Structured output over chat.** I prompt the model to return strict JSON in a fixed PRD schema, then render it into a real document UI. The user gets a usable artifact, not a wall of chat text — the difference between a toy and a tool.
- **Bring-your-own-key architecture.** I made a deliberate security call: rather than embed a key in a public static site (where anyone could drain it), the key lives only in the browser, with a demo mode so a cold visitor still sees real output. Judgment about deployment safety, not just model calls.
- **AI where it earns its keep.** The model does the heavy lifting on first-draft structure; the PM keeps the judgment. SpecCopilot removes the blank-page tax without pretending to replace product thinking — the same “where does AI actually add value” discernment I apply across my work.

### Impact

SpecCopilot collapses the slowest, most repetitive phase of spec-writing — the cold start — from a multi-hour drafting session into a single pass a PM refines from. More importantly, it demonstrates the operating model itself: I don't just specify AI features, I build working AI tools hands-on, make the architecture and safety calls myself, and ship them end-to-end. The same approach produced two other live products I built solo (AtithiPay, a multi-gateway payments platform; SmoothPaycheck, a launched B2C tool).

Try it live: [vimalajit11.github.io/spec-copilot](https://vimalajit11.github.io/spec-copilot) — open it, enter an idea, and watch it draft a PRD in one pass.

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